MIL-M-45976(WC)
19 January 1977

MILITARY SPECIFICATION

MOUNT, GUN: M140, M140A1, M150 AND M150A1

This specification is approved for use by the Rock Island Arsenal, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type hydrospring gun mount M140, M140A1, M150 and M150A1 to be mounted in combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal		
PPP- T-60	-	Tape, Packaging, Waterproof
Military		
MIL-P-116	-	Preservation-Packaging, Methods of
MIL-C-5501/7	-	Cap and Plug, Protective, Dust and Moisture Seal (Cap, Plug, General Purpose)
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Testing
MIL-V-13811	-	Varnish, Waterproofing, Electrical, Ignition
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Rock Island Arsenal, ATTN: SARRI-RLE-S, Rock Island, Illinois 61201 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1015

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MIL-B-26195	-	Box, Wood - Cleated, Skidded, Load-Bearing Base
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-45662	-	Calibration System Requirements
MIL-Q-45970	-	Quality Assurance for Weapons and Support Materiel
MIL-H-46170	-	Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydro-carbon Base
MIL-G-81322	-	Grease, Aircraft, General Purpose Wide Temperature Range

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods

DRAWINGS

Rock Island Arsenal

K10905448	-	Mount Assembly, Combination Gun: M140
K10940274	-	Mount Assembly, Combination Gun: M150
K12000890	-	Mount Assembly, Combination Gun: M140A1
K12000892	-	Mount Assembly, Combination Gun: M150A1
66F417	-	Machine, Test, Powder Type, Recoil Mech M140

(Product drawings referenced in this specification form a part of Drawing K10905448, K10940274, K12000890 or K12000892.

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PUBLICATIONS

Rock Island Arsenal

LQ10905448	-	Quality Requirements List: Mount, Gun: M140
LQ1094274	-	Quality Requirements List: Mount, Gun: M150
LQ12000890	-	Quality Requirements List: Mount, Gun: M140A1
LQ12000892	-	Quality Requirements List: Mount Gun: M150A1
EL10905448	-	List of Inspection Equipment Numbers: Mount, Gun: M140
EL10940274	-	List of Inspection Equipment Numbers: Mount, Gun: M150
EL12000890	-	List of Inspection Equipment Numbers: Mount, Gun: M140A1
EL12000892	-	List of Inspection Equipment Numbers: Mount, Gun: M150A1

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2).

3.2 Materials, construction and design. The mount shall conform to the materials, construction and design requirements specified herein and on Drawing K10905448, K10940274, K12000890 or K12000892, as applicable.

3.2.1 Hydraulic system. The hydraulic system shall be filled with hydraulic fluid conforming to MIL-H-46170. The initial and subsequent filling shall be filtered through a five micron filter until all air and moisture has been purged from the system.

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3.2.2 Fastener torque. Torque requirements for fasteners shall be as specified on the applicable drawing.

3.2.3 Lubrication. Lubrication requirements when assembling the mount shall be as specified on Drawing K10905448, K10940274, K12000890 or K12000892, as applicable.

3.3 Performance characteristics.

3.3.1 Static pressure. Prior to mechanical gymnastication and proof firing, the mount, with an adapter to hold the mount in battery, shall withstand the static pressure specified on Drawing K10905448, K10940274, K12000890 or K12000892, as applicable, for not less than five minutes. No drops of oil shall appear in the five minute pressure test. Caution: Extreme care shall be exercised during pressurization of the mount. Testing shall be as specified in 4.4 and 4.5.3.2.

3.3.2 Collar torque. Prior to mechanical gymnastication and proof firing, the collar (see Drawing 10915952) shall be torqued as specified on Drawing K10905448, K10940274, K12000890 or K12000892, as applicable. Testing shall be as specified in 4.4 and 4.5.3.3.

3.3.3 Mechanical gymnastication. The mount, including the gun or simulated equal, shall be gymnasticated 20 cycles at 0° elevation and 20 cycles at 35° elevation. The return cycle (counter recoil) commences at 12 inches from in-battery position. The return cycle time shall be measured on the last cycle at each elevation. The mount shall function at ambient temperature from 0° to plus 35° elevation without chatter or binding. Conditions such as erratic motion, hang-up during recoil, counter recoil or rebound upon return to battery shall not be acceptable. Testing shall be as specified in 4.4 and 4.5.3.4.

3.3.3.1 Return cycle time. The return cycle time for counter recoil during each mechanical gymnastication shall not exceed the requirements specified on Drawing K10905448, K10940274, K12000890 or K12000892 as applicable. Testing shall be as specified in 4.4 and 4.5.3.4.

3.3.3.2 Contaminated oil. Within five minutes after mechanical gymnastication, a minimum of one pint of oil shall be removed from the lowest drain hole in the cradle into a straight side glass container. Particles in the pint sample shall not exceed the limits specified on Drawing K10905448, K10940274, K12000890 or K12000892, as applicable. Testing shall be as specified in 4.4 and 4.5.3.4.

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3.3.3.3 Oil leakage. During and immediately after mechanical gymnastication, the oil leakage from a recoil mechanism shall be not more than 15 drops in a three minute period. After the mount has been allowed to set for four hours, no more than three drops of oil shall appear within a five minute period. Testing shall be as specified in 4.4 and 4.5.3.4.

3.3.4 Proof firing. The mount shall be capable of withstanding proof firing as specified herein without broken parts, hanging out-of-battery, erratic operation or other damage.

3.3.4.1 M140 and M140A1 mounts. The M140 and M140A1 mounts shall withstand proof firing in accordance with Table I (first article only) at each of the following ambient temperatures:

a. plus 70°F
b. plus 125°F
c. Minus 50°F

The mount shall be conditioned to the ambient temperature required prior to proof firing. Testing shall be as specified in 4.4 and 4.5.3.5.

TABLE I

PROOF FIRING

Round No.	Percent of Maximum Rated Powder Pressure	Firing Elevation (Degrees)
1	100	0
2	113	0
3	113	10
4	100	10
5	113	20
6	100	35
7	113	35
8	113	35
9	100	0

3.3.4.1.1 Recoil and counter recoil velocity. Counter recoil velocity, recoil travel, velocity and cylinder oil pressure versus time shall be as specified on Drawing K10905448 or K12000890, as applicable. Testing shall be as specified in 4.4.

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3.3.4.1.2 Breech operation. During counter recoil the breech operating cam shall operate in unison with the breech mechanism to eject cartridge cases clear of the gun and latch the breechlock in the open position. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.1.3 Recoil travel. Recoil travel for the M140 and M140A1 mounts shall be as specified on Drawing K10905448 or K12000890, as applicable. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.1.4 Total cycle time. Total cycle time (recoil and counter recoil) for the M140 and M140A1 mounts shall be as specified on Drawing K10905448 or K12000890, as applicable. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.1.5 Oil leakage. During and immediately after proof firing, the oil leakage from a recoil mechanism shall be not more than 15 drops in a three minute period. After the mount has been allowed to set for four hours, no more than three drops of oil shall appear within a five minute period. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.1.6 Ammunition. When proof firing the M140 and M140A1 mounts use HEAT-T, M456 cartridges or ballistically matched equal.

3.3.4.2 M140 and M140A1 mounts (powder gymnastication). Proof firing of the M140 and M140A1 mounts may be accomplished by powder gymnastication, subject to qualification of the contractor and approval by the contracting officer (see 6.2). The powder gymnasticator shall be in accordance with Drawing 66F417 or equivalent system. Powder gymnastication proof firing shall be as specified in Table II, conducted at prevailing ambient temperatures without conditioning. Only powder charges of 9, 10 and 11 ounces shall be fired. Testing shall be as specified in 4.5.3.6.

3.3.4.2.1 Recoil travel. Recoil travel for the M140 and M140A1 mounts shall be as specified on Drawing K10905448 or K12000890, as applicable. Testing shall be as specified in 4.5.3.6.

3.3.4.2.2 Total cycle time. Total cycle time (recoil and counter recoil) for the M140 and M140A1 mounts shall be as specified on Drawing K10905448 or K12000890, as applicable. Testing shall be as specified in 4.5.3.6.

TABLE II

POWDER HYMNASTICATION PROOF FIRING

Round No.	Peak Oil Pressure of M140 and M140A1 Mounts	Simulated Firing Elevation (Degrees)
1-15	2850-3200 PSI	35

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3.3.4.2.3 Oil leakage. During and immediately after powder gymnastication, the oil leakage from a recoil mechanism shall be not more than 15 drops in a three minute period. After the mount has been allowed to set for four hours, no more than three drops of oil shall appear within a five minute period. Testing shall be as specified in 4.5.3.6.

3.3.4.3 M150 and M150A1 mounts. The M150 and M150A1 mounts shall withstand proof firing in accordance with Table I (first article only), at each of the following ambient temperatures:

a. plus 70°F
b. plus 125°F
c. minus 50°F

The mount shall be conditioned to the ambient temperature required prior to proof firing. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.3.1 Recoil and counter recoil velocity. Counter recoil velocity, recoil travel, velocity and cylinder oil pressure versus time shall be as specified on Drawing K10940274 or K12000892, as applicable. Testing shall be as specified in 4.4.

3.3.4.3.2 Recoil travel. Recoil travel for the M150 and M150A1 mounts shall be as specified on Drawing K10940274 or K12000892, as applicable. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.3.3 Total cycle time. Total cycle time (recoil and counter recoil) for the M150 and M150A1 mounts shall be as specified on Drawing K10940274 or K12000892, as applicable. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.3.4 Oil leakage. During and immediately after proof firing, the oil leakage from a recoil mechanism shall be not more than 15 drops in a three minute period. After the mount has been allowed to set for four hours, no more than three drops of oil shall appear within a five minute period. Testing shall be as specified in 4.4 and 4.5.3.5.

3.3.4.3.5 Ammunition. When proof firing the M150 and M150A1 mounts use HEP, M123A1 cartridges or ballistically matched equal.

3.3.5 Durability. The mount shall be capable of withstanding live firing of 600 rounds with no evidence of structural damage, permanent deformation or failure. Ammunition shall be HEAT-T, M456 cartridges or ballistically matched equal. Testing shall be as specified in 4.5.3.7.

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3.4 Marking and identification. Marking and identification for the gun mount assembly shall be as specified on the applicable drawing and in accordance with MIL-STD-130. Serial numbers for the mounts shall be assigned by the procuring activity (see 6.2).

3.5 Workmanship. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations and mission operations which may affect serviceability, functioning, operation, appearance or safety. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair welding) or other similar practices shall not be permitted without prior approval of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall perform, as a minimum, inspections in accordance with the specifications, quality assurance provisions, MIL-Q-45970 and such documents as included in LQ10905448, LQ10940274, LQ12000890 or LQ12000892, as applicable, and the contract. These minimum inspections shall not be construed as relieving the contractor of his responsibilities under terms of the contract to furnish the Government with items conforming to the requirements of the contract, drawings and specifications.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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4.4 First article inspection. Three first article (initial production units) shall be submitted for inspection in accordance with the contract (see 6.2). The first article units shall be representative of the production processes to be used during quantity production. The first article mounts shall be subjected to the quality conformance inspection specified herein (except for 4.5.3.5, 4.5.3.6 and 4.5.3.7, which do not apply to first article), the proof firing testing specified in 4.4.1 and such other inspections as necessary to determine that all requirements of the contract have been met.

4.4.1 Proof firing testing (first article).

4.4.1.1 M140 and M140A1 mounts. Each mount shall be proof fired to determine compliance with 3.3.4, 3.3.4.1 and Table I. During proof firing, each mount shall be tested for recoil and counter recoil velocity (see 3.3.4.1.1), breech operation (see 3.3.4.1.2), recoil travel (see 3.3.4.1.3), total cycle time (see 3.3.4.1.4) and oil leakage (see 3.3.4.1.5). Upon completion of proof firing, each mount shall be tested for oil leakage (see 3.3.4.1.5). The test method shall be as specified in 4.6.4. Failure of any mount to meet the requirements shall be cause for rejection.

4.4.1.2 M150 and M150A1 mounts. Each mount shall be proof fired to determine compliance with 3.3.4, 3.3.4.3 and Table I. During proof firing, each mount shall be tested for recoil and counter recoil velocity (see 3.3.4.3.1), recoil travel (see 3.3.4.3.2), total cycle time (see 3.3.4.3.3) and oil leakage (see 3.3.4.3.4). Upon completion of proof firing, each mount shall be tested for oil leakage (see 3.3.4.3.4). The test method shall be as specified in 4.6.4. Failure of any mount to meet the requirements shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Mounts shall be assembled from component parts that have met all inspection requirements specified herein.

4.5.2 Examination. Visually and manually examine each mount to determine compliance with the applicable requirements. Each step in the examination shall include a visual examination for proper cleaning, the presence of the specified protective coating and to determine the general quality, completeness of manufacture and assembly and workmanship (see 3.5).

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4.5.2.1 Visually examine all markings required by the final assembly drawing to assure that they are clear and legible. Visually examine the identification plate for compliance with the applicable drawing, for secure assembly, serialization and to assure that all pertinent data has been applied (see 3.4).

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete mount. If test requirements cited herein are not met, acceptance of the mount shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine mounts, partially assembled mounts and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned mount for retest.

4.5.3.2 Static pressure testing. Prior to mechanical gymnastication and proof firing, each mount shall be static pressure tested to determine compliance with 3.3.1, using the test method specified in 4.6.1. Failure of the mount to meet the requirements shall be cause for rejection.

4.5.3.3 Collar torque testing. Prior to mechanical gymnastication and proof firing, the collar on each mount shall be checked for proper torque and locking to determine compliance with 3.3.2, using the test method specified in 4.6.2. Failure of the mount to meet the requirements shall be cause for rejection.

4.5.3.4 Mechanical gymnastication testing. Each mount shall be mechanically gymnasticated to determine compliance with 3.3.3, using the test method specified in 4.6.3. Failure of the mount to meet the requirements shall be cause for rejection.

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4.5.3.5 Proof firing testing.

4.5.3.5.1 M140 and M140A1 mounts. Each mount shall be proof fired to determine compliance with 3.3.4 and 3.3.4.1, using the test method specified in 4.6.4. The proof firing sequence shall be as specified in Table III, conducted at prevailing ambient temperature, without conditioning. During proof firing, each mount shall be tested for breech operation (see 3.3.4.1.2), recoil travel (see 3.3.4.1.3), total cycle time (see 3.3.4.1.4) and oil leakage (see 3.3.4.1.5). Upon completion of proof firing, each mount shall be tested for oil leakage (see 3.3.4.1.5). Failure of the mount to meet the requirements when proof firing as specified in Table III shall be cause for rejection. Proof firing of the M140 and M140A1 mounts may be accomplished by powder gymnastication (see 4.5.3.6).

4.5.3.5.2 M150 and M150A1 mounts. Each mount shall be proof fired to determine compliance with 3.3.4 and 3.3.4.3, using the test method specified in 4.6.4. The proof firing sequence shall be as specified in Table III, conducted at prevailing ambient temperature, without conditioning. During proof firing, each mount shall be tested for recoil travel (see 3.3.4.3.2), total cycle time (see 3.3.4.3.3) and oil leakage (see 3.3.4.3.4). Upon completion of proof firing, each mount shall be tested for oil leakage (see 3.3.4.3.4). Failure of the mount to meet the requirements when proof firing as specified in Table III shall be cause for rejection.

TABLE III

PROOF FIRING

Round No.	Percent of Maximum Rated Powder Pressure	Firing Elevation (Degrees)
1	100	0
2	113 ± 4	0
3	113 ± 4	20
4	113 ± 4	35
5	100	0

4.5.3.6 Powder gymnastication testing (M140 and M140A1 mounts only). Proof firing of the M140 and M140A1 mounts may be accomplished by powder gymnastication to determine compliance with 3.3.4.2, using the test method specified in 4.6.5 subject to qualification of the contractor and approval by the contracting officer (see 6.2). Contractor qualification for use of the powder gymnasticator for proof firing shall consist of subjecting the first 70 mounts produced, to a live five-round proof firing test in accordance with 4.5.3.5.1. Durability testing (see 4.5.3.7) may be substituted for contractor qualification. Failure of any mount to

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attain peak pressure as specified in Table II, recoil travel length (see 3.3.4.2.1) and total cycle time (see 3.3.4.2.2) on four consecutive cycles shall be considered a failure and shall be cause for rejection of the mount. Failure of any mount to meet the oil leakage requirements specified in 3.3.4.2.3 shall be cause for rejection. In addition, M140 and M140A1 mounts acceptance on the powder gymnasticator shall consist of the following criteria:

a. Lot sizes of 25 serialized mounts shall be gymnasticated in accordance with Table II.
b. One mount from each lot of 25 mounts shall be randomly selected and subjected to a live five round proof firing test in accordance with 4.5.3.5.1 at a Government proving ground.
c. Upon rejection of the one mount submitted for live proof firing, five additional mounts shall be randomly selected from the lot of 25 and shall also undergo the live five round proof firing test.
d. Rejection of any of the five additional mounts shall be cause for rejection of the entire lot.

4.5.3.7 Durability testing. To determine compliance with 3.3.5, the governments's representative shall randomly select one mount from the first lot of 30 mounts produced. The selected mount shall be subjected to a 600 round live firing test at a Government proving ground, using the test method specified in 4.6.6. The remaining 29 mounts shall be subjected to a live five round proof firing test in accordance with 4.5.3.5 at a Government proving ground. Failure of any mount shall be cause for rejection of the lot. Durability testing may be used as a substitute for contractor qualification for use of the powder gymnasticator.

4.5.4 Inspection equipment. Unless otherwise specified (see 6.2), responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment prescribed on List of Inspection Equipment Numbers EL10905448, EL10940274, EL12000890 or EL12000892, as applicable, and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.5 Inspection of packaging. Unless otherwise specified (see 6.2), inspection to determine compliance with preservation, packing and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-P-14232.

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4.6 Test methods.

4.6.1 Static pressure test. The mount, with an adapter to hold the mount in battery, shall be subjected to the specified static pressure for not less than five minutes. Visually examine for oil leakage (see 3.3.1).

4.6.2 Collar torque test. The mount shall be tested for proper collar torque and locking (see 3.3.2).

4.6.3 Mechanical gymnastication test. The mount and cannon assembly shall be mounted on a gymnaticator and the following shall be accomplished (see 3.3.3):

a. Set the travel limit switch contacts to the maximum recoil travel limits established for the particular mount being tested.
b. Elevate the mount until the gun is at 35° elevation, gymnasticate 20 times by moving the gun rearward the distance indicated in step a. above and then releasing it to return to the in-battery position under power of the recoil spring. On the last cycle, the mount shall be checked for time required to return to the in-battery position as specified in 3.3.3.1.
c. Depress the mount until the gun is at zero degrees elevation, gymnasticate 20 times as in b. above. On the last cycle, the mount shall be checked for time required to return to the in-battery position as specified in 3.3.3.1.
d. During gymnastication the mount shall be checked for oil leakage (see 3.3.3.3). Upon completion of gymnastication, the mount shall be checked for contaminated oil (see 3.3.3.2) and oil leakage (see 3.3.3.3).

4.6.4 Proof firing test. Proof firing of the mount shall be accomplished by assembling the mount and cannon tube to a vehicle, or a test stand simulating vehicle application. Proof firing shall be performed by the Government.

4.6.5 Powder gymnastication test. Proof firing of the mount by powder gymnastication shall be performed by the Government. The powder gymnasticator shall be in accordance with Drawing 66F417 or equivalent system, using powder charges of 9, 10 or 11 ounces.

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4.6.6 Durability test. The durability test shall be accomplished with the mount and cannon tube assembled to a vehicle, or a test stand simulated vehicle application. Firing shall be conducted at a Government proving ground. The first mount selected shall be fired 600 rounds and examined for structural damage, permanent deformation or failure. The remaining 29 mounts shall be proof fired as specified in 4.6.4.

5. PACKAGING

5.1 Disassembly. Disassembly of mounts shall be accomplished only as necessary to safeguard parts from damage, and provide required reduction in cube. Wherever practical, bolts, nuts, pins and washers shall be reinstalled on one of the mating parts and secured.

5.2 Preservation (Level A and B).

5.2.1 Cleaning and drying. Mounts shall be cleaned and dried in accordance with the applicable process and procedures of MIL-P-116.

5.2.1.1 Hydraulic components. Hydraulic components shall be cleaned in accordance with the applicable drawing.

5.2.2 Preservative application. All electrical items shall be preserved with compound conforming to MIL-V-13811. All exposed unpainted metal surfaces shall be coated with preservative conforming to type P-11 of MIL-P-116 or MIL-G-81322. All hydraulic components shall be preserved with oil type preservative conforming to MIL-H-6083.

5.2.3 Unit pack. All electrical contacts and openings of electrical items shall be covered with tape conforming to PPP-T-60, Type III, Class 2. All hydraulic parts and other openings shall be sealed with caps/plugs conforming to MIL-C-5501/7.

5.3 Packing (Level A and B). Mounts shall be placed in a wood box conforming to MIL-B-26195, Type II, Class and style optional. Plywood super structure is required. Blocking, bracing and cushioning shall be in accordance with MIL-STD-1186. Unit containers shall serve as the shipping containers.

5.4 Marking. Marking shall be in accordance with MIL-STD-129. "Reusable container" and "Center of balance" are markings required on each shipping container.

5.4.1 Special marking for hydraulic components. Hydraulic components shall be marked "DO NOT OPEN UNTIL READY FOR USE. CLEANED FOR HYDRAULIC SERVICE. IF INTERIOR IS CONTAMINATED, REPROCESS IN ACCORDANCE WITH DRAWING REQUIREMENTS."

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5.5 Repair parts. Repair parts shall be preserved, packed and marked in accordance with the Packaging Data Sheet or other requirements for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The M140, M140A1, M150 and M150A1 mounts covered by this specification are intended for use in combat vehicles.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the gun mount, showing applicable revision dates.
c. Shipping instructions for first article (see 3.1 and 4.4).
d. Approval for use of powder gymnastication (when applicable) (see 3.3.4.2 and 4.5.3.6).
e. Block of serial numbers (see 3.4).
f. Responsibility for acquisition, calibration, maintenance and disposition of inspection and test equipment, if other than as specified (see 4.5.4).
g. Packaging inspection, if other than specified (see 4.5.5).
h. Packaging instructions for repair parts (see 5.5).

Custodian:
Army - WC

Preparing Activity:
Army - WC

Project No.
1015-A264

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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